

LunarCrater-Net: Tile Resolution as a Key Factor in Automated Lunar Crater Detection via YOLOv8

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Abstract

Automated lunar crater detection is critical as accurate crater maps are essential for safe landing site selection and surface navigation. We present LunarCrater-Net, an automated crater detection pipeline using YOLOv8n fine-tuned via transfer learning on the Robbins (2018) lunar crater catalog. Our key finding is that reducing tile angular resolution from 60° to 15° per tile significantly improves detection from $mAP_{50}=0.0149$ to 0.481 , by increasing normalized crater box size from 0.003 to 0.022 . These results suggest that tile resolution is a critical and often overlooked parameter in planetary small object detection, with implications beyond lunar crater mapping to other bodies in the Solar System.

1 Introduction

Lunar impact craters are geological formations created by meteorite impacts on the Moon’s surface. Accurate crater maps are critical for space mission planning, including landing site selection for programs such as NASA’s Artemis and ISRO’s future lunar missions. Manual crater counting from orbital imagery is slow, subjective, and impossible to scale across the entire lunar surface. Automated detection methods are therefore essential for producing reliable, comprehensive crater catalogs.

Silburt et al. (2019) demonstrated the viability of convolutional neural networks for automated crater detection, achieving 92% recall on known craters using a CNN trained from scratch on lunar digital elevation maps. However, training from scratch requires large datasets and significant compute. In contrast, modern object detection frameworks with transfer learning offer a more accessible and reproducible approach.

In this work we present LunarCrater-Net, an automated crater detection pipeline using YOLOv8n with transfer learning from COCO weights. We demonstrate that reducing tile angular resolution from 60° to 15° per tile significantly improves detection from $\text{mAP50}=0.0149$ to 0.481 . Our key finding is that tile resolution is the dominant factor in small object detection performance for planetary crater mapping.

2 Related Work

Silburt et al. (2019) demonstrated a model trained via CNN from scratch on lunar DEMs achieving 92% recall on known craters. However, training from scratch requires large datasets and significant computations making it hard to reproduce or adapt to other planetary bodies. In contrast, LunarCrater-Net uses YOLOv8 with transfer learning, which outputs bounding boxes with coordinates instead of just detection masks.

3 Dataset

The Robbins (2018) lunar crater catalog contains 1.3 million craters mapped from NASA’s Lunar Reconnaissance Orbiter imagery. We filter to craters with diameter ≥ 5 km, as WAC imagery at 100m/px cannot reliably resolve smaller craters, yielding 83,061 craters.

Image tiles of 224×224 pixels covering 15×15 degrees of lunar surface are generated synthetically from catalog coordinates. Each crater is annotated in YOLO format: class x_{center} y_{center} width height, all normalized to $[0,1]$. The dataset is split 70% train / 20% validation / 10% test across 500 tiles.

4 Methodology

YOLOv8n is a lightweight object detection model with 3.2M parameters, capable of predicting bounding boxes and class labels in a single forward pass. It is pretrained on the COCO dataset (80 classes) and fine-tuned on our crater dataset, allowing the model to reuse learned visual features while learning crater-specific patterns.

The key parameter is `tile_deg` — reducing from 60° to 15° increases the normalized box size of a 10 km crater from 0.003 to 0.022, making craters significantly more visible to the model.

5 Results

We evaluate three model configurations progressively:

Reducing `tile_deg` from 60° to 15° produced a $32\times$ improvement in mAP50 . The baseline model failed because crater boxes of size 0.003 were too small for

Table 1: Training configuration

Parameter	Value
Optimizer	SGD (default Ultralytics)
Image size	224×224
Batch size	16
Epochs	100 (Phase 4)
Early stopping patience	20

Table 2: Model performance across phases

Model	Dataset	mAP50	mAP50-95	Epochs
YOLOv8n baseline	Synthetic, tile_deg=60	0.0149	0.0085	17
YOLOv8n improved	Synthetic, tile_deg=15	0.481	0.152	50
YOLOv8n real data	Real Robbins, tile_deg=15	0.298	0.124	100

the model to localize. Switching to real Robbins data reduced mAP50 to 0.298 despite 100 epochs, as real crater clustering patterns are harder to learn than uniform synthetic distributions.

6 Discussion

Our results demonstrate that reducing tile coverage from 60° to 15° increased crater box size from 0.003 to 0.022 — the single biggest improvement factor. This suggests that reducing tile resolution is the dominant factor in small object detection performance for planetary imagery.

The primary limitation is that our synthetic data overestimated real-world performance — mAP50 dropped from 0.481 to 0.298 when switching to the real Robbins catalog. Future work should incorporate real LROC WAC imagery instead of synthetic tiles, upgrade to YOLOv8s for higher model capacity, and evaluate performance separately on small, medium, and large craters.

7 Conclusion

We presented LunarCrater-Net, demonstrating that tile resolution is a critical and underexplored parameter in planetary crater detection. A single parameter change produced a 32× improvement in mAP50. Real catalog data proves harder than synthetic, highlighting the importance of domain-realistic training data.

References

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